

Full Length Article

Orbit-based universal approximation via hypercyclicity on compact–open topology

Farzona Mukhamedova^{a,*}, Ivan Tyukin^a, Farrukh Mukhamedov^{b,*}^a Department of Mathematics, King's College London, Strand, London, WC2R 2LS, United Kingdom^b Department of Mathematical Sciences, College of Science, United Arab Emirates University, P.O. Box 15551, Al Ain, Abu Dhabi, UAE

ARTICLE INFO

Keywords:

Universal approximation
Compact–open topology
Chaos
Hypercyclic vector
Dense orbit

ABSTRACT

We establish compact–open topology uniform universal approximation driven by hypercyclic orbits of an affine composition operator. In one dimension, for suitable continuous activation functions, we show that a fixed weighted translation operator is hypercyclic on the closure, in the compact–open topology, of the class of functions represented by one-hidden-layer neural networks. Consequently, there exists a single seed function in this closure whose iterates approximate every target uniformly on compact sets. We also prove density of periodic points, and hence chaoticity of the operator. Under the corresponding compact-wise density assumption for deeper and higher-dimensional network classes, the same orbit-based approximation mechanism extends to arbitrary depth and dimension. Thus approximation is achieved by moving along the orbit of one fixed seed under one fixed operator, rather than by re-optimizing parameters for each target.

1. Introduction

Classical universal approximation theorems, such as those by Cybenko (1989), Hornik, Stinchcombe and White (1989), Hornik (1991), and Funahashi and Nakamura (1993), show that shallow feedforward networks with suitable activation functions approximate continuous target functions on compact sets. For continuous non-polynomial activations, the standard characterization is due to Leshno et al. (1993). Most existing theories focus on approximation by static linear combinations of activation functions which require a new set of parameters for each target function.

Typically, approximating each new target function involves retraining the network parameters from scratch, a process that can be computationally expensive and theoretically repetitive. Additionally, many classical results rely on compactness of the input set (Barron (1991), Cybenko (1989), Hornik (1991), Hornik et al. (1989), Leshno et al. (1993), Long et al. (2007)). Most of the results have been obtained for continuous functions defined on compact spaces. Furthermore, van Nuland and Teun (2024) proved a non-compact universal approximation theorem for activations that are continuous and non-polynomial and asymptotically linear, establishing density in the vanishing at infinity space $C_0(\mathbb{R}^n)$. His analysis crucially exploits the linear growth of the

activation function φ at infinity, ensuring that network outputs do not explode.

Nevertheless, these results lack a mechanism for function approximation where the representation remains fixed, and adaptability arises from the action of a transformation. Therefore we focus on universal approximation in the compact–open topology where we are motivated by the following question. We ask whether one can construct a seed function in the compact–open closure of a neural-network class such that, without changing the underlying network parameters, its orbit under a fixed transformation can approximate many target functions on compact sets.

Our work fills this gap by proposing the framework where universal approximation arises from the dense orbit of a single function under the composition of an operator which we show is hypercyclic. Consequently, this reduces the need for repeatedly retraining network parameters. In light of this, our findings clarify how approximation of continuous functions can be generated by the orbit of a single function belonging to the relevant network closure. In practice, this may reduce the computational cost since rather than exploring the full space, we can work within the countable dense orbit so approximation boils down to choosing a single iterate index of the operator. We leave some possible applications to families of integral and differential equations are left for future work.

* Corresponding authors.

E-mail addresses: farzona.mukhamedova@kcl.ac.uk (F. Mukhamedova), far75m@gmail.com (F. Mukhamedov).

1.1. Function spaces and the compact–open topology

In what follows we shall work with several topologies on spaces of continuous functions. We work over the field $\mathbb{R}$ and we let $\mathbb{N} = \{1, 2, \dots\}$. For $n \in \mathbb{N}$, let $C(\mathbb{R}^n)$ denote the space of all continuous functions $f : \mathbb{R}^n \rightarrow \mathbb{R}$. We write $C_b(\mathbb{R}^n)$ for the bounded continuous functions, $C_0(\mathbb{R}^n)$ for the continuous functions vanishing at infinity (i.e., $\lim_{\|x\| \rightarrow \infty} f(x) = 0$), and $C_c(\mathbb{R}^n)$ for the compactly supported continuous functions. Explicitly,

$$C_0(\mathbb{R}^n) := \{f \in C(\mathbb{R}^n) : \forall \varepsilon > 0 \exists R > 0 \text{ such that } |f(x)| < \varepsilon \text{ whenever } \|x\| \geq R\}.$$

For a compact set $K \subset \mathbb{R}^n$, define the seminorm as

$$p_K(f) := \sup_{x \in K} |f(x)|.$$

The compact–open topology on $C(\mathbb{R}^n)$ is the locally convex topology generated by the family $\{p_K : K \subset \mathbb{R}^n \text{ compact}\}$. A sequence $\{f_m\}$ converges to f in this topology if and only if $p_K(f_m - f) \rightarrow 0$ for every compact K , equivalently, if and only if $f_m \rightarrow f$ uniformly on every compact set.

Because $\mathbb{R}^n$ is σ -compact, the compact–open topology is metrizable. For example, if $K_m := [-m, m]^n$, then

$$d_{co}(f, g) := \sum_{m=1}^{\infty} 2^{-m} \min\{1, p_{K_m}(f - g)\} \quad (1.1)$$

induces the compact–open topology on $C(\mathbb{R}^n)$. The factor $\min\{1, \cdot\}$ keeps each summand bounded by 2^{-m} and therefore guarantees convergence of the series while preserving the same notion of compactwise convergence. We write co for compact–open closure.

For a subset $S \subset C(\mathbb{R}^n)$, we write $\overline{S}^{co}$ for its closure in the compact–open topology. Given $A, B \subset \mathbb{R}^n$, we also use

$$\text{dist}(A, B) := \inf\{\|x - y\| : x \in A, y \in B\}$$

for the Euclidean distance between subsets of $\mathbb{R}^n$.

Intuitively, one can imagine that the compact–open topology forgets what happens far away from a given compact set and this flexibility is exactly what allows us to construct hypercyclic orbits for weighted translations, whereas hypercyclicity is impossible in the uniform norm on $C_0(\mathbb{R}^n)$.

The classical cut-off arguments show that $C_c(\mathbb{R}^n)$ is dense in $C(\mathbb{R}^n)$ for the compact–open topology, and hence $C_0(\mathbb{R}^n)$ is dense in $C(\mathbb{R}^n)$ in this sense. Later we exploit this density together with the weaker nature of the compact–open topology to construct hypercyclic vectors lying in $C_0(\mathbb{R}^n)$ whose orbits are dense in the compact–open closures of the corresponding network classes which are defined below.

1.2. Weighted translation operator

Hypercyclicity of an operator and its hypercyclic vector have long been studied as a form of universality in the field of dynamical systems. This shows the existence of a vector with a dense orbit under a set operator and this has been studied as a form of universality in the field of dynamical systems. The monographs Bayart and Étienne Matheron (2009), Grosse-Erdmann et al. (2011) illustrate, for instance, that certain weighted shift operators on sequence spaces are hypercyclic. This hints at a dynamic analogue of neural universal approximation and rather than varying neural network parameters across tasks, one might iterate a fixed operator to sweep out a dense set of functions.

We recall the following definitions from Grosse-Erdmann et al. (2011) which will be useful in what follows.

Definition 1.1. Let X be a separable topological vector space and let $T : X \rightarrow X$ be a continuous linear operator. An operator $T : X \rightarrow X$ is called hypercyclic if there exists $x \in X$ whose orbit $\{T^n x : n \geq 0\}$ under T is dense in X . In that case, x is called a hypercyclic vector for T .

A point $x \in X$ is called periodic for T if $T^p x = x$ for some integer $p \geq 1$.

Definition 1.2. A linear operator T on X is said to be chaotic if it is hypercyclic and has a dense set of periodic points.

The main dynamical operator in this paper is the weighted translation operator. Let $\lambda \in \mathbb{R} \setminus \{0\}$ be a non-zero scalar (weight) and $\beta \in \mathbb{R}$. The (weighted) translation operator $T_{\lambda, \beta}$ is defined by,

$$(T_{\lambda, \beta} f)(x) = \lambda f(x + \beta) \quad f \in C(\mathbb{R}).$$

Iterating this operator $T_{\lambda, \beta}$ gives rise to a semigroup $\{T_{\lambda, \beta}^k\}$ of operators acting on function spaces. One can infer that,

$$(T_{\lambda, \beta}^k f)(x) = \lambda^k f(x + k\beta), \quad k \in \mathbb{N}. \quad (1.2)$$

The central role of the operator is to shift arguments of f which we use to study orbit behavior of functions under repeated application of $T_{\lambda, \beta}$. We denote the identity operator on the ambient function space as Id . The orbit of f under $T_{\lambda, \beta}$ is the set $\{T_{\lambda, \beta}^k f : k \geq 0\}$, where $T_{\lambda, \beta}^0 = \text{Id}$. In particular, we are interested in whether the orbit $\{T_{\lambda, \beta}^k f\}$ is dense in the relevant compact–open closure of a network class. In other words, we are checking for hypercyclicity as defined in definition 1.1. In what follows, we are going to employ methods of hypercyclic operators from Bayart and Étienne Matheron (2009) to construct universal approximators inside compact–open closures of neural-network function classes.

1.3. Neural network function spaces

The support of a function f is denoted by $\text{supp } f$. Let φ be a non-polynomial, continuous activation function on $\mathbb{R}$.

We define the space of one-layer feed-forward neural networks with n input nodes and one output node as the following subspace of $C(\mathbb{R}^n)$:

$$\mathcal{N}_{\varphi}^1(\mathbb{R}^n) := \text{span}\{x \mapsto \varphi(a \cdot x + b) : a \in \mathbb{R}^n, b \in \mathbb{R}\}, \quad (1.3)$$

where for $a, x \in \mathbb{R}^n$, we denote by $a \cdot x := \sum_{j=1}^n a_j x_j$ the Euclidean inner product. Equivalently, $\mathcal{N}_{\varphi}^1(\mathbb{R}^n)$ can be written as follows

$$\mathcal{N}_{\varphi}^1(\mathbb{R}^n) := \left\{ \sum_{i=1}^m c_i \varphi(a_i \cdot x + b_i) : m \in \mathbb{N}, a_i \in \mathbb{R}^n, b_i, c_i \in \mathbb{R} \right\}, \quad (1.4)$$

The space of networks with depth ℓ ($\ell \geq 2$), which means that we have ℓ hidden layers, can then be defined recursively (see Fig. 1).

$$\mathcal{N}_{\varphi}^{\ell}(\mathbb{R}^n) := \text{span}\{x \mapsto \varphi(f(x) + b) : f \in \mathcal{N}_{\varphi}^{\ell-1}(\mathbb{R}^n), b \in \mathbb{R}\}. \quad (1.5)$$

We define,

$$X_1 := \overline{\mathcal{N}_{\varphi}^1(\mathbb{R}^n)}^{co},$$

where the superscript co indicates that the closure is taken in the compact–open topology.

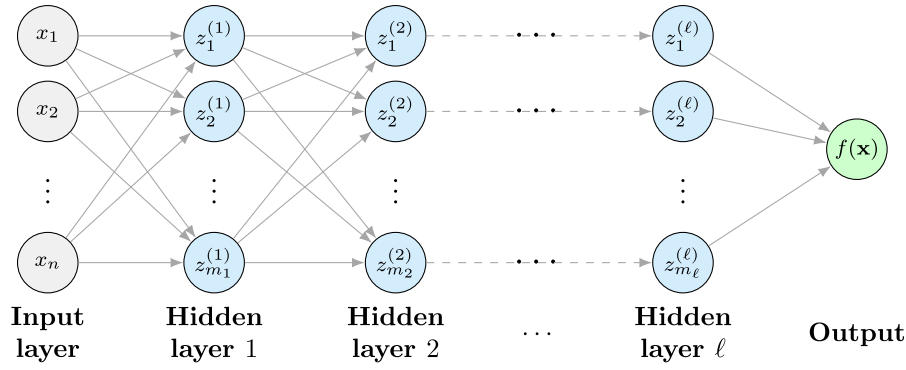
We recall that for a compact $K \subset \mathbb{R}^n$, by $C(K)$ we denote the set of all continuous functions with the norm $\|f\|_{C(K)} = \max_{x \in K} |f(x)|$.

Lemma 1. Let $\mathcal{F} \subset C(\mathbb{R}^n)$. Assume that for every compact $K \subset \mathbb{R}^n$, the restricted set $\mathcal{F}|_K := \{f|_K : f \in \mathcal{F}\}$ is dense in $C(K)$ with respect to $\|\cdot\|_{C(K)}$. Then $\overline{\mathcal{F}}^{co} = C(\mathbb{R}^n)$. In particular, $C_0(\mathbb{R}^n) \subset \overline{\mathcal{F}}^{co}$.

Proof. Let $g \in C(\mathbb{R}^n)$ and let U be a basic neighborhood of g in the compact–open topology. Then U has the form

$$U = \bigcap_{i=1}^m \{h \in C(\mathbb{R}^n) : \|h - g\|_{C(K_i)} < \varepsilon_i\},$$

for some compacts K_i and $\varepsilon_i > 0$. Put $K := \bigcup_{i=1}^m K_i$ (still compact) and $\varepsilon := \min_i \varepsilon_i$. By assumption, choose $f \in \mathcal{F}$ such that $\|f - g\|_{C(K)} < \varepsilon$. Then for each i we have $\|f - g\|_{C(K_i)} \leq \|f - g\|_{C(K)} < \varepsilon \leq \varepsilon_i$, so $f \in U$. Hence every neighborhood of g meets $\mathcal{F}$, i.e. $g \in \overline{\mathcal{F}}^{co}$. $\square$



$$z^{(0)} = \mathbf{x}, \quad z_j^{(s)} = \varphi \left(\sum_i w_{ji}^{(s)} z_i^{(s-1)} + b_j^{(s)} \right), \quad 1 \leq s \leq \ell,$$

$$f(\mathbf{x}) = \sum_{j=1}^{m_\ell} c_j z_j^{(\ell)}.$$

Fig. 1. A depth- ℓ feedforward neural network on $\mathbb{R}^n$. The hidden layers apply the activation function φ , and the output is a finite linear combination of the last hidden-layer activations.

Using the last lemma, it is enough to prove that for every compact $K \subset \mathbb{R}^n$, the restricted class $\mathcal{N}_\varphi^\ell(\mathbb{R}^n)|_K$ is dense in $C(K)$. For $\ell = 1$, this is the classical theorem of Leshno et al. [Leshno et al. \(1993\)](#). For $\ell \geq 2$, we argue by induction on ℓ .

Assume that $\mathcal{N}_\varphi^{\ell-1}(\mathbb{R}^n)|_K$ is dense in $C(K)$ for every compact $K \subset \mathbb{R}^n$. Let $g \in C(K)$ and $\varepsilon > 0$. By the case $\ell = 1$, choose

$$G(\mathbf{x}) = \sum_{r=1}^m c_r \varphi(\mathbf{a}_r \cdot \mathbf{x} + b_r) \in \mathcal{N}_\varphi^1(\mathbb{R}^n)$$

such that

$$\|g - G\|_{C(K)} < \frac{\varepsilon}{2}.$$

Since each affine map $\mathbf{x} \mapsto \mathbf{a}_r \cdot \mathbf{x}$ is continuous on K , and φ is uniformly continuous on compact intervals, there exists $\delta > 0$ such that whenever

$$\|h_r - (\mathbf{a}_r \cdot \mathbf{x})\|_{C(K)} < \delta \quad (r = 1, \dots, m),$$

one has

$$\left\| \sum_{r=1}^m c_r \varphi(h_r + b_r) - \sum_{r=1}^m c_r \varphi(\mathbf{a}_r \cdot \mathbf{x} + b_r) \right\|_{C(K)} < \frac{\varepsilon}{2}.$$

By the induction hypothesis, for each r there exists $h_r \in \mathcal{N}_\varphi^{\ell-1}(\mathbb{R}^n)$ such that

$$\|h_r - (\mathbf{a}_r \cdot \mathbf{x})\|_{C(K)} < \delta.$$

Hence

$$H(\mathbf{x}) := \sum_{r=1}^m c_r \varphi(h_r(\mathbf{x}) + b_r) \in \mathcal{N}_\varphi^\ell(\mathbb{R}^n),$$

and

$$\|H - g\|_{C(K)} \leq \|H - G\|_{C(K)} + \|G - g\|_{C(K)} < \varepsilon.$$

Thus $\mathcal{N}_\varphi^\ell(\mathbb{R}^n)|_K$ is dense in $C(K)$ for every $\ell \in \mathbb{N}$. Therefore, by [Lemma 1](#),

$$\overline{\mathcal{N}_\varphi^\ell(\mathbb{R}^n)}^{co} = C(\mathbb{R}^n), \quad C_0(\mathbb{R}^n) \subset \overline{\mathcal{N}_\varphi^\ell(\mathbb{R}^n)}^{co}.$$

We are going to show the existence of a single hypercyclic function F within the compact-open closure $\overline{\mathcal{N}_\varphi^1(\mathbb{R}^n)}^{co}$. Its orbit under repeated application of $T_{\lambda, \beta}$ is dense in X_1 , and since $C_0(\mathbb{R}) \subset X_1$, every target

in $C_0(\mathbb{R})$ can be approximated on compact sets by suitable iterates of this orbit. The function itself need not be a single finite network, but rather, it lies in the compact-open closure of the finite-network class and can therefore be approximated on each compact set by finite neural networks with parameters fixed once those approximants are chosen. The same mechanism will be used and shown that it holds for higher-depth closures.

1.4. Expressive power approximation

Once a hypercyclic function $F \in \overline{\mathcal{N}_\varphi^1(\mathbb{R})}^{co}$ is constructed with respect to $T_{\lambda, \beta}$ it can approximate any target function $g \in C_0(\mathbb{R})$ not by retraining its parameters, but by iterating the operator $T_{\lambda, \beta}$ applied to F . Hence, for any $g \in C_0(\mathbb{R})$, any compact $K \subset \mathbb{R}$, and any tolerance $\varepsilon > 0$, there exists an integer $k \in \mathbb{N}$ such that,

$$\|T_{\lambda, \beta}^k F - g\|_{C(K)} < \varepsilon. \quad (1.6)$$

Equivalently, the orbit $\{T_{\lambda, \beta}^n F : n \geq 0\}$ is dense in $X_1 := \overline{\mathcal{N}_\varphi^1(\mathbb{R})}^{co}$ for the compact-open topology. Since $C_0(\mathbb{R}) \subset X_1$, every target in $C_0(\mathbb{R})$ can be approximated uniformly on compact sets by selecting a suitable iterate. This shows that orbit-based universal approximation is achieved by navigating along a dense orbit rather than by re-optimizing network parameters.

In this subsection, “expressive power” refers to the size and approximation capability of the function class generated by a given neural-network architecture. A complementary line of research studies how increasing the depth of a network enlarges this realizable class. Depth-separation results show that there exist functions which can be represented efficiently by deep networks, but whose representation, or approximation to comparable accuracy, by shallower networks requires substantially larger width [Hanin and Rolnick \(2019\)](#), [Hanin and Sellke \(2017\)](#), [Montúfar et al. \(2014\)](#), [Raghu et al. \(2017\)](#), [Telgarsky \(2016\)](#), [Yarotsky \(2017\)](#). More precisely, “exponentially larger width” means that, for certain target functions, reducing the depth forces the number of neurons per layer to grow like e^{cm} , or equivalently 2^{cm} , in a relevant complexity parameter m , whereas the deeper network realizes or approximates the same function with only polynomial, or even linear,

width. In parallel, sharp approximation rates have been obtained for deep ReLU networks on Sobolev-type function classes [Yarotsky \(2017\)](#). Further refinements emphasize that hierarchical or compositional structure in the target function can be exploited by deep architectures to achieve more efficient approximation [Lin et al. \(2017\)](#), [Petersen and Voigtlaender \(2018\)](#). Additional perspectives on the expressive advantages of depth and the limitations of shallow architectures can be found in [Eldan and Shamir \(2016\)](#), [Liang and Srikant \(2016\)](#), [Lu et al. \(2017\)](#).

1.5. Related work

The roots of approximation theory go back to classical density results such as Weierstrass's theorem, which states that every continuous function on a compact interval can be uniformly approximated by polynomials [Weierstrass \(1885\)](#). This was later generalized by Stone's theorem, which gives conditions under which subalgebras are dense in $C(K)$ for compact spaces K [Stone \(1948\)](#). Another important classical result is Kolmogorov's superposition theorem, which shows that continuous multivariate functions can be represented through finite superpositions of univariate functions [Kolmogorov \(1957\)](#). In a different direction, operator-theoretic notions such as hypercyclicity and supercyclicity have also been studied over non-Archimedean spaces [Mukhamedov and Khakimov \(2018\)](#), [Mukhamedov et al. \(2022\)](#).

The passage from classical approximation theory to neural networks was marked by universal approximation theorems, which show that neural networks with suitable activation functions can approximate broad classes of target functions. These qualitative approximation results were later refined into quantitative theories describing approximation rates and complexity. In particular, Fourier-analytic methods led to the Barron norm and to dimension-free approximation estimates for certain sigmoidal networks [Barron \(2002\)](#). At the same time, lower-bound results revealed intrinsic limitations of width-restricted architectures [Maiorov and Pinkus \(1999\)](#). These developments form part of the broader capacity theory of neural networks [Pinkus \(1999\)](#).

A related line of work studies approximation under architectural restrictions. For instance, fixed-weight constructions can still achieve near-optimal error bounds in suitable regimes [Guliyev and Ismailov \(2018\)](#). Other works compare greedy and random choices of basis functions in approximation schemes [Gorban et al. \(2016\)](#). Related high-dimensional phenomena have also been investigated outside the neural-network setting. For example, replacing the Euclidean norm by fractional quasi-norms can mitigate distance concentration in some heavy-tailed settings, but such improvements are not universal [Tyukin et al. \(2025\)](#). These results indicate that randomization or a change of metric alone does not provide a general guarantee of expressive power without additional structural assumptions.

Another, conceptually different, direction concerns quantum-like models of cognition and their possible coupling to neuronal dynamics. In that literature, the abbreviation GPT stands for *generalized probabilistic theory*, not for generative pre-trained transformer. A generalized probabilistic theory is an abstract operational framework in which states form a convex subset of an ordered real vector space. The order is usually determined by a positive cone, which specifies which states and effects are physically or probabilistically admissible. Probabilities are then obtained by pairing states with effects. Thus, an ordered-state-space GPT model provides a common mathematical framework that includes classical probability models and quantum probability models as special cases. Such models have been used to describe non-classical behavioral phenomena in cognition, including contextuality, interference effects, and deviations from classical laws of total probability. Recent works have explored how these ordered-state-space models can be coupled to neuronal networks, with the aim of connecting behavioral-level quantum-like effects to neural-level state representations and decoherence-based decision dynamics [Asano et al. \(2011\)](#), [Khrennikov et al. \(2018, 2025\)](#).

1.6. Motivation and overview of results

Almost every universal approximation theorem, whether on compact or non-compact sets, relies on tailoring the weights to the target function. As a consequence, the network must be retrained or fine tuning each time the task changes. Hypercyclic operator theory [Bayart and Étienne Matheron \(2009\)](#), [Grosse-Erdmann et al. \(2011\)](#) suggests an alternative where we keep the network fixed and approximate new targets by a suitable weighted translation operator.

Naturally this raises some questions: can one find a function network whose orbit under repeated composition is dense in a prescribed function space, and what additional structure does such an orbit possess?

The paper answers these questions and establishes several results.

- We prove the existence of a function $F \in \overline{\mathcal{N}_\varphi^1}^{co}$ whose orbit under the operator $T_{\lambda,\beta}$ the sequence $\{T_{\lambda,\beta}^n F : n \geq 0\}$ is dense in $C_0(\mathbb{R})$ for the compact-open topology whenever $\beta \neq 0$. This provides the first hypercyclic-orbit realization of non-compact universal approximation within a standard neural network class.
- We prove that the composition operator $T_{\lambda,\beta}$ is hypercyclic and chaotic. Due to the chaotic property it means that the weighted translation operator admits a network whose iterates are able to hit any targets on compact sets. Moreover, periodic points of the induced operator are dense in the compact-open closure X_1 but nearby seeds can diverge after iterates.
- For every depth $\ell \in \mathbb{N}$, we construct a hypercyclic vector F_ℓ . Thus, hypercyclic orbit-based approximation persists at every depth.
- Any desired target function $g \in C_0(\mathbb{R})$ can be approximated on compact subsets by choosing a suitable iterate n of the operator $T_{\lambda,\beta}$ applied to the fixed seed function F .
- Orbit-based approximation using hypercyclicity demonstrates that expressive power in neural networks can arise from the geometry of input orbits rather than from repeated optimization of weights.

1.7. Structure of the paper

The paper is organized as follows. In [Section 1](#) we recall several milestones in approximation theory and neural network expressivity. We highlight the progression from classical density theorems, to modern depth separation results and alternative constructions. We provide the preliminary information for the setting of our paper and outline the motivation and overview of our results. [Section 2](#) outlines the principal results for orbit based approximation on $\mathbb{R}$. [Section 3](#) establishes results for orbit based approximation and its analogues for neural networks at arbitrary depth and on $\mathbb{R}^n$. In [Section 4](#) we discuss the implications of our results, go through some open questions and further directions for practitioners as there is a lot of uncovered terrains that this work opens up from a theoretical and practical point of view. We emphasize that the proposed approach is novel in the setting considered here.

2. Orbit-based approximation on $\mathbb{R}$

Lemma 2. Let $\varphi \in C(\mathbb{R})$ and set $X_1 := \overline{\mathcal{N}_\varphi^1}^{co}$. For $\lambda \in \mathbb{R} \setminus \{0\}$ and $\beta \in \mathbb{R}$ define,

$$(T_{\lambda,\beta} f)(x) := \lambda f(x + \beta).$$

Then $T_{\lambda,\beta}(\mathcal{N}_\varphi^1(\mathbb{R})) \subset \mathcal{N}_\varphi^1(\mathbb{R})$, hence $T_{\lambda,\beta}(X_1) \subset X_1$. Moreover, $T_{\lambda,\beta} : X_1 \rightarrow X_1$ is linear and continuous for the compact-open topology.

Proof. For $f(x) = c \varphi(ax + b)$ we have

$$(T_{\lambda,\beta} f)(x) = \lambda c \varphi(a(x + \beta) + b) = \lambda c \varphi(ax + (a\beta + b)) \in \mathcal{N}_\varphi^1(\mathbb{R}).$$

Since finite sums are preserved, $T_{\lambda,\beta}(\mathcal{N}_\varphi^1(\mathbb{R})) \subset \mathcal{N}_\varphi^1(\mathbb{R})$. Linearity is immediate from the definition.

For continuity on $C(\mathbb{R})$ with the compact-open topology, for every compact $K \subset \mathbb{R}$,

$$\sup_{x \in K} |(T_{\lambda,\beta} f)(x) - (T_{\lambda,\beta} g)(x)| = |\lambda| \sup_{x \in K} |f(x + \beta) - g(x + \beta)|$$

$$= |\lambda| \|f - g\|_{C(K+\beta)},$$

and $K + \beta$ is compact. Equivalently, for the canonical seminorms $p_K(h) := \sup_{x \in K} |h(x)|$,

$$p_K(T_{\lambda,\beta} f) \leq |\lambda| p_{K+\beta}(f),$$

so $T_{\lambda,\beta}$ is continuous on $C(\mathbb{R})$, hence on the subspace X_1 .

Finally, since $T_{\lambda,\beta}$ is continuous and $T_{\lambda,\beta}(\mathcal{N}_\varphi^1(\mathbb{R})) \subset \mathcal{N}_\varphi^1(\mathbb{R})$, we get

$$T_{\lambda,\beta}(X_1) = T_{\lambda,\beta}(\overline{\mathcal{N}_\varphi^1(\mathbb{R})}^{co}) \subset \overline{T_{\lambda,\beta}(\mathcal{N}_\varphi^1(\mathbb{R}))}^{co} \subset \overline{\mathcal{N}_\varphi^1(\mathbb{R})}^{co} = X_1,$$

so X_1 is invariant under $T_{\lambda,\beta}$. $\square$

We notice that by the theorem of Michael [Michael \(1961\)](#) the space $X_1 = \overline{\mathcal{N}_\varphi^1(\mathbb{R})}^{co}$ is separable with respect to the compact-open topology. Therefore, it makes sense to ask whether $T_{\lambda,\beta} : X_1 \rightarrow X_1$ is hypercyclic.

Theorem 2.1. *Assume that $\varphi \in C(\mathbb{R})$ is non-polynomial. Let $\beta \in \mathbb{R} \setminus \{0\}$ and $\lambda \in \mathbb{R} \setminus \{0\}$. Let $X_1 = \overline{\mathcal{N}_\varphi^1(\mathbb{R})}^{co}$, then the operator $T_{\lambda,\beta}$ is hypercyclic on X_1 for the compact-open topology.*

Proof. By [Lemma 2](#), $T_{\lambda,\beta}$ is a continuous linear operator on X_1 . Moreover, since φ is continuous and non-polynomial, the classical universal approximation theorem implies that $\mathcal{N}_\varphi^1(\mathbb{R})$ is dense in $C(K)$ for every compact interval $K \subset \mathbb{R}$. Hence, by [Lemma 1](#), we infer that $X_1 = C(\mathbb{R})$ with the compact-open topology. Therefore it is enough to construct a function $F \in C(\mathbb{R})$ whose orbit under $T_{\lambda,\beta}$ is dense in $C(\mathbb{R})$ for the compact-open topology.

Let $\mathcal{D}$ be the countable family of pairs (I, q) such that $I = [a, b]$ is a compact interval with rational endpoints $a < b$, and q is a continuous piecewise affine function on I with rational breakpoints and rational values at those breakpoints. This family is dense in the following cofinal sense: for every compact set $K \subset \mathbb{R}$, every $G \in C(\mathbb{R})$, and every $\delta > 0$, there exists $(I, q) \in \mathcal{D}$ such that

$$K \subset I \quad \text{and} \quad \|q - G\|_{C(K)} < \delta.$$

Indeed, one first chooses a rational interval I containing K , then uses uniform continuity of G on I to approximate $G|_I$ by a piecewise affine function, and finally perturbs the breakpoints and nodal values slightly so that they are rational.

Choose an enumeration

$$(K_m, f_m)_{m \geq 1}$$

of $\mathcal{D}$ in which every element of $\mathcal{D}$ occurs infinitely many times. Consequently, this enumeration is tail-dense: for every compact $K \subset \mathbb{R}$, every $G \in C(\mathbb{R})$, every $\delta > 0$, and every $M \in \mathbb{N}$, there exists $m \geq M$ such that

$$K \subset K_m \quad \text{and} \quad \|f_m - G\|_{C(K)} < \delta.$$

For each $m \geq 1$, by the compact universal approximation property, choose $g_m \in \mathcal{N}_\varphi^1(\mathbb{R})$ such that

$$\|g_m - f_m\|_{C(K_m)} < 2^{-m}.$$

We now place these local targets far apart along the translation direction. Since $\beta \neq 0$, for each fixed compact interval K_m the translates $K_m + n\beta$ escape every compact subset of $\mathbb{R}$ as $n \rightarrow \infty$. We may therefore choose integers

$$1 \leq n_1 < n_2 < \dots$$

inductively so that, with

$$E_m := K_m + n_m \beta,$$

the following two conditions hold:

$$\text{dist}(E_m, [-m, m]) > 2,$$

and

$$\text{dist}(E_m, E_j) > 4 \quad \text{whenever } j < m.$$

For each m , set

$$U_m := \{x \in \mathbb{R} : \text{dist}(x, E_m) < 1\}.$$

Then the open sets U_m are pairwise disjoint and locally finite. Indeed, if $K \subset [-R, R]$ is compact, then for all $m \geq R$ we have $K \subset [-m, m]$, and therefore $U_m \cap K = \emptyset$ by the choice of E_m .

For each m , choose a continuous cutoff function $\chi_m \in C_c(\mathbb{R})$ such that

$$0 \leq \chi_m \leq 1, \quad \chi_m \equiv 1 \text{ on } E_m, \quad \text{supp } \chi_m \subset U_m.$$

The supports of the functions χ_m are therefore pairwise disjoint and locally finite.

Define

$$h_m(x) := \lambda^{-n_m} g_m(x - n_m \beta) \chi_m(x),$$

and set

$$F(x) := \sum_{m=1}^{\infty} h_m(x).$$

Since the family $\{\text{supp } \chi_m : m \geq 1\}$ is locally finite, this series is locally finite. Hence $F \in C(\mathbb{R})$. Since $X_1 = C(\mathbb{R})$, we have $F \in X_1$.

We claim that the orbit of F is dense in X_1 . Fix $m \geq 1$ and $x \in K_m$. Then

$$x + n_m \beta \in E_m.$$

Therefore $\chi_m(x + n_m \beta) = 1$. Moreover, since the sets U_j are pairwise disjoint and $\text{supp } \chi_j \subset U_j$, we have

$$x + n_m \beta \notin \text{supp } \chi_j \quad \text{for } j \neq m.$$

Thus only the m -th block contributes at the point $x + n_m \beta$, and hence

$$\begin{aligned} (T_{\lambda,\beta}^{n_m} F)(x) &= \lambda^{n_m} F(x + n_m \beta) \\ &= \lambda^{n_m} h_m(x + n_m \beta) \\ &= \lambda^{n_m} \lambda^{-n_m} g_m((x + n_m \beta) - n_m \beta) \chi_m(x + n_m \beta) \\ &= g_m(x). \end{aligned}$$

Consequently,

$$(T_{\lambda,\beta}^{n_m} F)|_{K_m} = g_m|_{K_m}.$$

Now let $G \in X_1$, let $K \subset \mathbb{R}$ be compact, and let $\varepsilon > 0$. Since $X_1 = C(\mathbb{R})$, we may regard G as an element of $C(\mathbb{R})$. Choose $M \in \mathbb{N}$ so large that

$$2^{-M} < \frac{\varepsilon}{2}.$$

By the tail-density of the enumeration $(K_m, f_m)_{m \geq 1}$, there exists $m \geq M$ such that

$$K \subset K_m \quad \text{and} \quad \|f_m - G\|_{C(K)} < \frac{\varepsilon}{2}.$$

Using the exact interpolation identity above and the fact that $K \subset K_m$, we obtain

$$\begin{aligned} \|T_{\lambda,\beta}^{n_m} F - G\|_{C(K)} &\leq \|T_{\lambda,\beta}^{n_m} F - g_m\|_{C(K)} + \|g_m - f_m\|_{C(K)} + \|f_m - G\|_{C(K)} \\ &= 0 + \|g_m - f_m\|_{C(K)} + \|f_m - G\|_{C(K)} \\ &\leq \|g_m - f_m\|_{C(K_m)} + \|f_m - G\|_{C(K)} \\ &< 2^{-m} + \frac{\varepsilon}{2} \\ &\leq 2^{-M} + \frac{\varepsilon}{2} < \varepsilon. \end{aligned}$$

Thus, for every $G \in X_1$, every compact $K \subset \mathbb{R}$, and every $\varepsilon > 0$, some iterate of F approximates G uniformly on K .

Finally, a basic neighborhood in the compact-open topology is determined by finitely many compact sets. Replacing those compact sets by their union reduces the general case to the estimate above. Hence the orbit

$$\{T_{\lambda,\beta}^n F : n \geq 0\}$$

is dense in X_1 . Therefore $T_{\lambda,\beta}$ is hypercyclic on X_1 . $\square$

Corollary 2.1. Assume that $\varphi \in C(\mathbb{R})$ is non-polynomial. Let $\beta \neq 0$ and let $\lambda \in \mathbb{R}$ satisfy $|\lambda| > 1$. Then the hypercyclic vector in Theorem 2.1 may be chosen to belong to $C_0(\mathbb{R})$. In particular, there exists $F \in C_0(\mathbb{R}) \cap X_1$ such that the orbit $\{T_{\lambda,\beta}^n F : n \geq 0\}$ is dense in X_1 for the compact-open topology.

Proof. We use the same construction as in Theorem 2.1, but choose the integers n_m with one additional amplitude condition.

For each m , let

$$L_m := \{y \in \mathbb{R} : \text{dist}(y, K_m) \leq 1\}.$$

Since g_m is continuous and L_m is compact, the quantity

$$A_m := \|g_m\|_{C(L_m)}$$

is finite. Because $|\lambda| > 1$, we may choose n_m sufficiently large so that, in addition to the separation conditions from the proof of Theorem 2.1, one has

$$|\lambda|^{-n_m} A_m \leq 2^{-m}.$$

After choosing n_m , set

$$E_m := K_m + n_m\beta, \quad U_m := \{x \in \mathbb{R} : \text{dist}(x, E_m) < 1\},$$

and choose $\chi_m \in C_c(\mathbb{R})$ satisfying

$$0 \leq \chi_m \leq 1, \quad \chi_m \equiv 1 \text{ on } E_m, \quad \text{supp } \chi_m \subset U_m.$$

Define

$$h_m(x) := \lambda^{-n_m} g_m(x - n_m\beta) \chi_m(x), \quad F(x) := \sum_{m=1}^{\infty} h_m(x).$$

As before, the supports are pairwise disjoint and locally finite, so $F \in C(\mathbb{R})$.

Moreover, if $x \in \text{supp } \chi_m$, then $x - n_m\beta \in L_m$. Hence

$$|h_m(x)| \leq |\lambda|^{-n_m} \|g_m\|_{C(L_m)} = |\lambda|^{-n_m} A_m \leq 2^{-m}.$$

Thus

$$\|h_m\|_{\infty} \leq 2^{-m}.$$

We now prove that $F \in C_0(\mathbb{R})$. Let $\varepsilon > 0$. Choose $M \in \mathbb{N}$ such that $2^{-M} < \varepsilon$.

Since the finitely many functions $h_1, \dots, h_{M-1}$ have compact support, there exists $R > 0$ such that

$$\text{supp } h_m \subset [-R, R] \quad \text{for } 1 \leq m < M.$$

If $|x| > R$, then only terms with $m \geq M$ can contribute. Since the supports are pairwise disjoint, at most one such term is nonzero at x . Therefore

$$|F(x)| \leq \sup_{m \geq M} \|h_m\|_{\infty} \leq \sup_{m \geq M} 2^{-m} = 2^{-M} < \varepsilon.$$

Hence $F(x) \rightarrow 0$ as $|x| \rightarrow \infty$, so $F \in C_0(\mathbb{R})$.

The interpolation identity

$$(T_{\lambda,\beta}^{n_m} F)|_{K_m} = g_m|_{K_m}$$

is unchanged. Therefore the same tail-density argument used in the proof of Theorem 2.1 shows that the orbit of F is dense in X_1 for the compact-open topology (see Fig. 2). $\square$

Therefore, Theorem 2.1 gives density of the orbit in

$$X_1 := \overline{\mathcal{N}_{\varphi}^1(\mathbb{R})}^{co}.$$

Since $C_0(\mathbb{R}) \subset X_1$, every target in $C_0(\mathbb{R})$ is approximable uniformly on compact subsets by the same orbit.

Therefore, Theorem 2.1 gives density of the orbit in X_1 . Since $C_0(\mathbb{R}) \subset X_1$, every target in $C_0(\mathbb{R})$ is approximable uniformly on compact subsets by the same orbit.

Remark 2.2. If $|\lambda| \leq 1$, then no function $F \in C_0(\mathbb{R})$ can be a hypercyclic vector for $T_{\lambda,\beta}$. Indeed, assume that $F \in C_0(\mathbb{R})$ and fix a compact K . Since $K + n\beta \rightarrow \infty$ and F vanishes at infinity,

$$\|T^n F\|_{C(K)} = |\lambda|^n \sup_{x \in K} |F(x + n\beta)| \xrightarrow{n \rightarrow \infty} 0.$$

Hence $T^n F \rightarrow 0$ in the compact-open topology, so the orbit cannot be dense.

Remark 2.3. In the uniform norm on $C_0(\mathbb{R})$, the operator $T_{\lambda,\beta}$ cannot be hypercyclic for any constant λ (the orbit either preserves, explodes, or collapses the norm). Indeed, if $|\lambda| < 1$ and $F \in C_0(\mathbb{R})$, then

$$\|T_{\lambda,\beta}^n F\|_{C(\mathbb{R})} = |\lambda|^n \|F\|_{C(\mathbb{R})} \xrightarrow{n \rightarrow \infty} 0,$$

so no such F can be hypercyclic. Now, assume that $|\lambda| > 1$, then $\|T_{\lambda,\beta}^n F\|_{C(\mathbb{R})} \xrightarrow{n \rightarrow \infty} \infty$. Hence, 0 function cannot be approximated.

On the other hand, the compact-open topology measures errors only on a fixed compact K , and our placement of the blocks forces all non- m terms to vanish on K at time n_m . This is exactly the mechanism the uniform norm forbids.

Proposition 2.4. Assume that $\varphi \in C(\mathbb{R})$ is non-polynomial. Let $\lambda \neq 0$, $\beta \neq 0$, and $k \in \mathbb{N}$. Let $I \subset \mathbb{R}$ be a bounded interval, and let $|I|$ denote the length of interval I . Assume that $|I| < k|\beta|$.

Choose $\psi \in C_c(\mathbb{R})$ such that $\text{supp } \psi \subset I^\circ$, where I° denotes the interior of I . Define

$$f(x) := \sum_{j \in \mathbb{Z}} \lambda^{-kj} \psi(x - jk\beta).$$

Then $f \in X_1$. Moreover,

$$T_{\lambda,\beta}^k f = f.$$

Thus, if $\psi \neq 0$, then f is a nonzero periodic point of $T_{\lambda,\beta}$, with period dividing k .

Proof. Since $|I| < k|\beta|$ and $\text{supp } \psi \subset I^\circ$, the translated supports

$$\text{supp } \psi + jk\beta, \quad j \in \mathbb{Z},$$

are pairwise disjoint. Indeed, if

$$y \in (\text{supp } \psi + jk\beta) \cap (\text{supp } \psi + mk\beta)$$

for some $j \neq m$, then there exist $x, z \in \text{supp } \psi \subset I$ such that

$$y = x + jk\beta = z + mk\beta.$$

Hence

$$x - z = (m - j)k\beta.$$

Therefore

$$|x - z| = |m - j|k|\beta| \geq k|\beta|,$$

which contradicts

$$|x - z| \leq |I| < k|\beta|.$$

Thus the translated supports are pairwise disjoint.

The family

$$\{\text{supp } \psi + jk\beta : j \in \mathbb{Z}\}$$

is also locally finite, because $\text{supp } \psi$ is compact and $\beta \neq 0$. Hence the series

$$f(x) = \sum_{j \in \mathbb{Z}} \lambda^{-kj} \psi(x - jk\beta)$$

is locally finite. Therefore f is well defined and continuous on $\mathbb{R}$.

We next prove the periodicity identity. For every $x \in \mathbb{R}$,

$$\begin{aligned} (T_{\lambda,\beta}^k f)(x) &= \lambda^k f(x + k\beta) \\ &= \lambda^k \sum_{j \in \mathbb{Z}} \lambda^{-kj} \psi(x + k\beta - jk\beta) \\ &= \lambda^k \sum_{j \in \mathbb{Z}} \lambda^{-kj} \psi(x - (j-1)k\beta). \end{aligned}$$

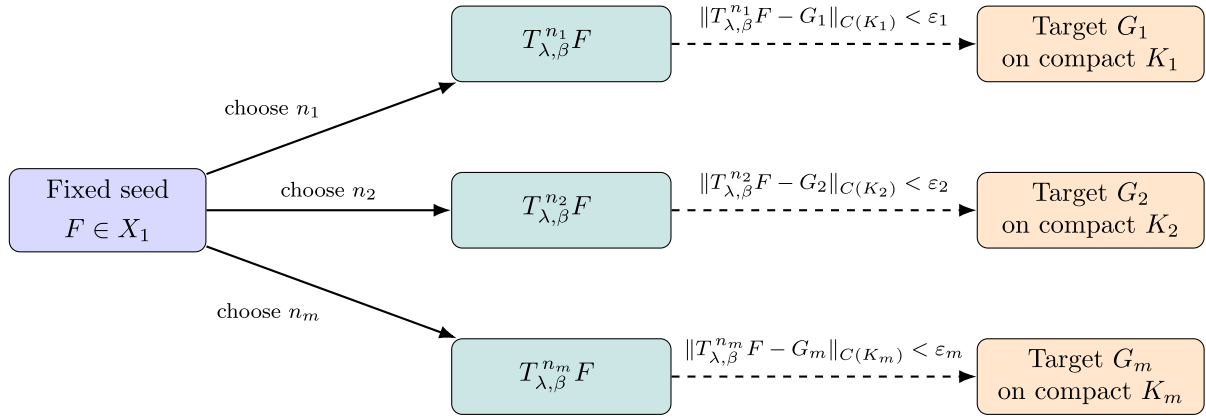


Fig. 2. Orbit-based approximation by a fixed seed. Note, one seed F and one operator $T_{\lambda,\beta}$ are fixed. Then function $F \in X_1$ generates approximants to different targets by applying different powers of the weighted translation operator $T_{\lambda,\beta}$.

Putting $i = j - 1$, we obtain

$$\begin{aligned} (T_{\lambda,\beta}^k f)(x) &= \lambda^k \sum_{i \in \mathbb{Z}} \lambda^{-k(i+1)} \psi(x - ik\beta) \\ &= \sum_{i \in \mathbb{Z}} \lambda^{-ki} \psi(x - ik\beta) \\ &= f(x). \end{aligned}$$

Therefore, one finds

$$T_{\lambda,\beta}^k f = f.$$

It remains to show that $f \in X_1$. Let $K \subset \mathbb{R}$ be compact and let $\varepsilon > 0$. Since the translated supports are locally finite, the set

$$J(K) := \{j \in \mathbb{Z} : K \cap (\text{supp } \psi + jk\beta) \neq \emptyset\}$$

is finite.

If $J(K) = \emptyset$, then $f = 0$ on K . Since $0 \in \mathcal{N}_\varphi^1(\mathbb{R})$, we immediately get $\|f - 0\|_{C(K)} = 0 < \varepsilon$.

Assume now that $J(K) \neq \emptyset$. Define the compact set

$$K' := \bigcup_{j \in J(K)} (K - jk\beta).$$

By the compact-wise density of $\mathcal{N}_\varphi^1(\mathbb{R})$, choose $g \in \mathcal{N}_\varphi^1(\mathbb{R})$ such that

$$\|\psi - g\|_{C(K')} < \delta,$$

where

$$\delta := \frac{\varepsilon}{\sum_{j \in J(K)} |\lambda|^{-kj}}.$$

Define

$$G_K(x) := \sum_{j \in J(K)} \lambda^{-kj} g(x - jk\beta).$$

Since $g \in \mathcal{N}_\varphi^1(\mathbb{R})$ and the one-hidden-layer network class is closed under translations of the input, scalar multiplication, and finite sums, we have $G_K \in \mathcal{N}_\varphi^1(\mathbb{R})$.

For $x \in K$, only indices in $J(K)$ can contribute to $f(x)$. Hence

$$f(x) = \sum_{j \in J(K)} \lambda^{-kj} \psi(x - jk\beta).$$

Therefore,

$$\begin{aligned} |f(x) - G_K(x)| &= \left| \sum_{j \in J(K)} \lambda^{-kj} (\psi(x - jk\beta) - g(x - jk\beta)) \right| \\ &\leq \sum_{j \in J(K)} |\lambda|^{-kj} \|\psi - g\|_{C(K')} \\ &< \delta \sum_{j \in J(K)} |\lambda|^{-kj} \\ &= \varepsilon. \end{aligned}$$

Hence

$$\|f - G_K\|_{C(K)} < \varepsilon$$

with $G_K \in \mathcal{N}_\varphi^1(\mathbb{R})$. Since K and ε were arbitrary, f belongs to the compact-open closure of $\mathcal{N}_\varphi^1(\mathbb{R})$. Hence $f \in X_1$. $\square$

Theorem 2.5. Assume that $\varphi \in C(\mathbb{R})$ is non-polynomial. Let $\beta \neq 0$ and $\lambda \neq 0$, then $T_{\lambda,\beta}$ is chaotic. More precisely:

- (i) $T_{\lambda,\beta}$ is hypercyclic on X_1 ;
- (ii) for every $k \in \mathbb{N}$ there exists a nonzero $f \in X_1$ such that

$$T_{\lambda,\beta}^k f = f,$$

and the set of periodic points of $T_{\lambda,\beta}$ is dense in X_1 .

Proof. Part (i) follows from Theorem 2.1, since $\beta \neq 0$ and $\lambda \neq 0$. It remains to prove that periodic points are dense in X_1 .

By Proposition 2.4 we know, for every $k \in \mathbb{N}$, there exists a nonzero $f \in X_1$ satisfying

$$T_{\lambda,\beta}^k f = f.$$

It remains to prove that the set of periodic points is dense in X_1 . Let $G \in X_1$ and $K \subset \mathbb{R}$ be compact. Take an arbitrary $\varepsilon > 0$. Since $X_1 = C(\mathbb{R})$, the function G is continuous. Choose $k \in \mathbb{N}$ so large that there exists a bounded interval $I \subset \mathbb{R}$ with

$$K \subset I^\circ \quad \text{and} \quad |I| < k|\beta|.$$

Choose $\eta \in C_c(\mathbb{R})$ such that

$$0 \leq \eta \leq 1, \quad \eta \equiv 1 \text{ on } K, \quad \text{supp } \eta \subset I^\circ.$$

Set

$$\psi := \eta G.$$

Then $\psi \in C_c(\mathbb{R})$ and $\text{supp } \psi \subset I^\circ$. Define

$$f(x) := \sum_{j \in \mathbb{Z}} \lambda^{-kj} \psi(x - jk\beta).$$

By the same argument as in the proof of Proposition 2.4, we infer that $f \in X_1$ and $T_{\lambda,\beta}^k f = f$. Thus f is a periodic point.

We claim that $f = G$ on K . Indeed, if $x \in K$, then $x \in I^\circ$. Since $|I| < k|\beta|$, the intervals

$$I + jk\beta, \quad j \in \mathbb{Z},$$

are pairwise disjoint. Therefore, for $x \in K \subset I^\circ$, the only term in the series for $f(x)$ that can contribute is the term with $j = 0$. Hence

$$f(x) = \psi(x) = \eta(x)G(x) = G(x), \quad x \in K.$$

Consequently,

$$\|f - G\|_{C(K)} = 0 < \varepsilon.$$

Since G , K , and ε were arbitrary, periodic points are dense in X_1 .

Combining hypercyclicity from part (i) with density of periodic points, we conclude that $T_{\lambda, \beta}$ is chaotic on X_1 . $\square$

Remark 2.6. If we consider the mapping

$$(Tf)(x) := \lambda f(\alpha x + \beta),$$

where $|\alpha| \neq 1$ and $\beta \neq 0$, then it is not hypercyclic. Indeed, the map $\tau(x) = \alpha x + \beta$ has a unique fixed point $x_* = -\beta/(\alpha - 1)$. For every $F \in X_1$ and $n \geq 1$,

$$(T^n_{\lambda, \beta} F)(x_*) = \lambda^n F(x_*).$$

Since evaluation at x_* is a continuous nonzero functional on X_1 , the image of any orbit under this functional is the geometric progression $\{\lambda^n F(x_*) : n \geq 0\}$, which is never dense in $\mathbb{R}$. Therefore T is not hypercyclic on X_1 , and hence not chaotic, for any $\lambda \neq 0$.

Theorem 2.7. Let $\varphi \in C(\mathbb{R})$ be non-polynomial. Set

$$X_\ell := \overline{\mathcal{N}_\varphi^\ell(\mathbb{R})}^{co},$$

where the closure is taken with respect to the compact-open topology. If $\beta \neq 0$ and $\lambda \neq 0$, then $T_{\lambda, \beta}$ is hypercyclic on X_ℓ .

Proof. We first check that $T_{\lambda, \beta}$ is a well-defined continuous linear operator on X_ℓ .

Let

$$S_\beta f(x) := f(x + \beta).$$

We claim that

$$S_\beta(\mathcal{N}_\varphi^\ell(\mathbb{R})) \subset \mathcal{N}_\varphi^\ell(\mathbb{R}).$$

For $\ell = 1$, this follows from

$$\varphi(a(x + \beta) + b) = \varphi(ax + (a\beta + b)).$$

Assume now that the claim holds for depth $\ell - 1$. A generator of $\mathcal{N}_\varphi^\ell(\mathbb{R})$ has the form

$$x \mapsto \varphi(h(x) + c), \quad h \in \mathcal{N}_\varphi^{\ell-1}(\mathbb{R}), \quad c \in \mathbb{R}.$$

Then

$$\varphi(h(x + \beta) + c) = \varphi((S_\beta h)(x) + c).$$

By the induction hypothesis,

$$S_\beta h \in \mathcal{N}_\varphi^{\ell-1}(\mathbb{R}).$$

Hence

$$x \mapsto \varphi(h(x + \beta) + c)$$

belongs to $\mathcal{N}_\varphi^\ell(\mathbb{R})$. Since $\mathcal{N}_\varphi^\ell(\mathbb{R})$ is a linear span, the claim follows.

Therefore

$$T_{\lambda, \beta}(\mathcal{N}_\varphi^\ell(\mathbb{R})) \subset \mathcal{N}_\varphi^\ell(\mathbb{R}).$$

Moreover, $T_{\lambda, \beta}$ is continuous in the compact-open topology because, for every compact interval $K \subset \mathbb{R}$,

$$\|T_{\lambda, \beta} f\|_{C(K)} = |\lambda| \|f\|_{C(K+\beta)}.$$

Since $K + \beta$ is compact, this proves compact-open continuity. Hence $T_{\lambda, \beta}(X_\ell) \subset X_\ell$.

By the non-polynomiality of φ and Lemma 1, we have $X_\ell = \overline{\mathcal{N}_\varphi^\ell(\mathbb{R})}^{co} = C(\mathbb{R})$ with the compact-open topology. Thus it remains to construct a function $F_\ell \in C(\mathbb{R})$ whose orbit under $T_{\lambda, \beta}$ is dense in $C(\mathbb{R})$ for the compact-open topology.

Let $\mathcal{D}$ be the countable family of pairs (I, q) such that $I = [a, b]$ is a compact interval with rational endpoints $a < b$, and q is a continuous piecewise affine function on I with rational breakpoints and rational values at those breakpoints. Enumerate $\mathcal{D}$ as $(K_m, f_m)_{m \geq 1}$ in such a way

that every pair occurs infinitely many times. Then this enumeration is tail-dense: for every compact interval $K \subset \mathbb{R}$, every $G \in C(\mathbb{R})$, every $\delta > 0$, and every $M \in \mathbb{N}$, there exists $m \geq M$ such that

$$K \subset K_m \quad \text{and} \quad \|f_m - G\|_{C(K)} < \delta.$$

For each $m \geq 1$, the compact-wise density of $\mathcal{N}_\varphi^\ell(\mathbb{R})$ gives a function $g_m \in \mathcal{N}_\varphi^\ell(\mathbb{R})$ such that

$$\|g_m - f_m\|_{C(K_m)} < 2^{-m}.$$

We now place these local targets far apart along the translation direction. Since $\beta \neq 0$, the translates $K_m + n\beta$ escape every compact subset of $\mathbb{R}$ as $n \rightarrow \infty$. Choose integers

$$1 \leq n_1 < n_2 < \dots$$

inductively so that, with

$$E_m := K_m + n_m \beta,$$

we have

$$\text{dist}(E_m, [-m, m]) > 2$$

and

$$\text{dist}(E_m, E_j) > 4 \quad \text{whenever } j < m.$$

For each m , define

$$U_m := \{x \in \mathbb{R} : \text{dist}(x, E_m) < 1\}.$$

Then the sets U_m are pairwise disjoint and locally finite.

Choose $\chi_m \in C_c(\mathbb{R})$ such that

$$0 \leq \chi_m \leq 1, \quad \chi_m \equiv 1 \text{ on } E_m, \quad \text{supp } \chi_m \subset U_m.$$

Define

$$h_m(x) := \lambda^{-n_m} g_m(x - n_m \beta) \chi_m(x),$$

and set

$$F_\ell(x) := \sum_{m=1}^{\infty} h_m(x).$$

Because the family $(U_m)_{m \geq 1}$ is locally finite, the above sum is locally finite. Hence $F_\ell \in C(\mathbb{R}) = X_\ell$.

We next prove the interpolation identity. Fix $m \geq 1$ and $x \in K_m$. Then $x + n_m \beta \in E_m$. Therefore $\chi_m(x + n_m \beta) = 1$. Moreover, since the supports of the χ_j are contained in pairwise disjoint sets U_j , no block with $j \neq m$ contributes at $x + n_m \beta$. Thus

$$\begin{aligned} (T_{\lambda, \beta}^{n_m} F_\ell)(x) &= \lambda^{n_m} F_\ell(x + n_m \beta) \\ &= \lambda^{n_m} h_m(x + n_m \beta) \\ &= \lambda^{n_m} \lambda^{-n_m} g_m((x + n_m \beta) - n_m \beta) \chi_m(x + n_m \beta) \\ &= g_m(x). \end{aligned}$$

Hence

$$(T_{\lambda, \beta}^{n_m} F_\ell)|_{K_m} = g_m|_{K_m}.$$

We now show that the orbit of F_ℓ is dense. Let $G \in X_\ell = C(\mathbb{R})$, let $K \subset \mathbb{R}$ be compact, and let $\varepsilon > 0$. Choose $M \in \mathbb{N}$ so large that

$$2^{-M} < \frac{\varepsilon}{2}.$$

By tail-density, choose $m \geq M$ such that

$$K \subset K_m \quad \text{and} \quad \|f_m - G\|_{C(K)} < \frac{\varepsilon}{2}.$$

Then

$$\begin{aligned} \|T_{\lambda, \beta}^{n_m} F_\ell - G\|_{C(K)} &\leq \|T_{\lambda, \beta}^{n_m} F_\ell - g_m\|_{C(K)} + \|g_m - f_m\|_{C(K)} + \|f_m - G\|_{C(K)} \\ &= 0 + \|g_m - f_m\|_{C(K)} + \|f_m - G\|_{C(K)} \\ &\leq \|g_m - f_m\|_{C(K_m)} + \|f_m - G\|_{C(K)} \\ &< 2^{-m} + \frac{\varepsilon}{2} \\ &\leq 2^{-M} + \frac{\varepsilon}{2} \\ &< \varepsilon. \end{aligned}$$

Thus, for every $G \in X_\ell$, every compact $K \subset \mathbb{R}$, and every $\varepsilon > 0$, some iterate of F_ℓ approximates G uniformly on K .

Finally, a basic neighborhood in the compact-open topology is determined by finitely many compact sets. Replacing those compact sets by their union reduces the general case to the estimate above. Therefore the orbit

$$\{T_{\lambda, \beta}^n F_\ell : n \geq 0\}$$

is dense in X_ℓ . Hence $T_{\lambda, \beta}$ is hypercyclic on X_ℓ . $\square$

3. Hypercyclicity at arbitrary depth and on $\mathbb{R}^n$

Throughout this section, let $\varphi \in C(\mathbb{R})$ be non-polynomial. Let us recall the necessary definitions.

Definition 3.1. Let $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ be an activation function and let $n \in \mathbb{N}$. For $\mathbf{x}, \mathbf{a} \in \mathbb{R}^n$, write

$$\mathbf{a} \cdot \mathbf{x} := \sum_{i=1}^n a_i x_i$$

for the Euclidean inner product.

The class of one-hidden-layer φ -networks on $\mathbb{R}^n$ is defined by

$$\mathcal{N}_\varphi^1(\mathbb{R}^n) = \left\{ \sum_{j=1}^m c_j \varphi(\mathbf{a}_j \cdot \mathbf{x} + b_j) : m \in \mathbb{N}, c_j, b_j \in \mathbb{R}, \mathbf{a}_j \in \mathbb{R}^n \right\}.$$

For $\ell \geq 2$, the class of depth- ℓ φ -networks is defined recursively by

$$\mathcal{N}_\varphi^\ell(\mathbb{R}^n) = \left\{ \sum_{j=1}^m c_j \varphi(g_j(\mathbf{x}) + b_j) : m \in \mathbb{N}, c_j, b_j \in \mathbb{R}, g_j \in \mathcal{N}_\varphi^{\ell-1}(\mathbb{R}^n) \right\}.$$

Here span always means the set of all finite real linear combinations.

We write

$$X_\ell^n := \overline{\mathcal{N}_\varphi^\ell(\mathbb{R}^n)}^{co},$$

where the closure is taken with respect to the compact-open topology, that is, uniform convergence on compact subsets of $\mathbb{R}^n$. For $n = 1$ we write $X_\ell := X_\ell^1$.

Since the activation function φ is non-polynomial, then by Leshno [Leshno et al. \(1993\)](#) together with the Induction argument, we obtain

$$\mathcal{N}_\varphi^\ell(\mathbb{R}^n)|_K \text{ is dense in } C(K) \quad (3.1)$$

for every compact set $K \subset \mathbb{R}^n$. By the compact-wise density lemma, this implies $X_\ell^n = C(\mathbb{R}^n)$ with the compact-open topology.

For $\lambda \in \mathbb{R}$ and $\mathbf{b} \in \mathbb{R}^n$, define the operator

$$(T_{\lambda, \mathbf{b}} f)(\mathbf{x}) := \lambda f(\mathbf{x} + \mathbf{b}), \quad f : \mathbb{R}^n \rightarrow \mathbb{R}.$$

Lemma 3. Let $\lambda \in \mathbb{R}$ and $\mathbf{b} \in \mathbb{R}^n$. Then

$$T_{\lambda, \mathbf{b}}(\mathcal{N}_\varphi^\ell(\mathbb{R}^n)) \subset \mathcal{N}_\varphi^\ell(\mathbb{R}^n).$$

Consequently, $T_{\lambda, \mathbf{b}}(X_\ell^n) \subset X_\ell^n$. Moreover, $T_{\lambda, \mathbf{b}} : X_\ell^n \rightarrow X_\ell^n$ is linear and continuous for the compact-open topology.

Proof. Linearity is immediate. We first prove the invariance of $\mathcal{N}_\varphi^\ell(\mathbb{R}^n)$ under translations. Let

$$S_{\mathbf{b}} f(\mathbf{x}) := f(\mathbf{x} + \mathbf{b}).$$

For $\ell = 1$, if

$$f(\mathbf{x}) = \sum_{r=1}^m c_r \varphi(\mathbf{a}_r \cdot \mathbf{x} + d_r),$$

then

$$(S_{\mathbf{b}} f)(\mathbf{x}) = \sum_{r=1}^m c_r \varphi(\mathbf{a}_r \cdot (\mathbf{x} + \mathbf{b}) + d_r) = \sum_{r=1}^m c_r \varphi(\mathbf{a}_r \cdot \mathbf{x} + (\mathbf{a}_r \cdot \mathbf{b} + d_r)).$$

Thus

$$S_{\mathbf{b}} f \in \mathcal{N}_\varphi^1(\mathbb{R}^n).$$

Assume now that translation invariance holds at depth $\ell - 1$. A generator of $\mathcal{N}_\varphi^\ell(\mathbb{R}^n)$ has the form

$$\mathbf{x} \mapsto \varphi(h(\mathbf{x}) + d), \quad h \in \mathcal{N}_\varphi^{\ell-1}(\mathbb{R}^n), \quad d \in \mathbb{R}.$$

After translation we obtain

$$\varphi(h(\mathbf{x} + \mathbf{b}) + d) = \varphi((S_{\mathbf{b}} h)(\mathbf{x}) + d).$$

By the induction hypothesis, $S_{\mathbf{b}} h \in \mathcal{N}_\varphi^{\ell-1}(\mathbb{R}^n)$. Hence the translated generator belongs to $\mathcal{N}_\varphi^\ell(\mathbb{R}^n)$. Since $\mathcal{N}_\varphi^\ell(\mathbb{R}^n)$ is a linear span, it is invariant under translations. Multiplication by the scalar λ also preserves the class. Therefore $T_{\lambda, \mathbf{b}}(\mathcal{N}_\varphi^\ell(\mathbb{R}^n)) \subset \mathcal{N}_\varphi^\ell(\mathbb{R}^n)$.

Next, for every compact set $K \subset \mathbb{R}^n$,

$$\|T_{\lambda, \mathbf{b}} f - T_{\lambda, \mathbf{b}} g\|_{C(K)} = |\lambda| \|f - g\|_{C(K+\mathbf{b})}.$$

Since $K + \mathbf{b}$ is compact, this proves continuity of $T_{\lambda, \mathbf{b}}$ in the compact-open topology. Hence, by continuity and the invariance of the network class,

$$T_{\lambda, \mathbf{b}}(X_\ell^n) = T_{\lambda, \mathbf{b}}\left(\overline{\mathcal{N}_\varphi^\ell(\mathbb{R}^n)}^{co}\right) \subset \overline{T_{\lambda, \mathbf{b}}(\mathcal{N}_\varphi^\ell(\mathbb{R}^n))}^{co} \subset X_\ell^n.$$

$\square$

Theorem 3.1. let $\varphi \in C(\mathbb{R})$ be non-polynomial, and $\lambda \neq 0$ and $\mathbf{b} \in \mathbb{R}^n \setminus \{0\}$. Then $T_{\lambda, \mathbf{b}}$ is hypercyclic for the compact-open topology.

Proof. By [Lemma 3](#), $T_{\lambda, \mathbf{b}}$ is a continuous linear operator on X_ℓ^n . Moreover, we have $X_\ell^n = C(\mathbb{R}^n)$ with the compact-open topology. It is therefore enough to construct a function $F \in C(\mathbb{R}^n)$ whose orbit under $T_{\lambda, \mathbf{b}}$ is dense in $C(\mathbb{R}^n)$ for the compact-open topology.

Let $\mathcal{Q}$ be the countable family of pairs (Q, p) such that Q is a compact box in $\mathbb{R}^n$ with rational endpoints and p is a polynomial with rational coefficients. Enumerate $\mathcal{Q}$ as $(Q_m, p_m)_{m \geq 1}$ in such a way that every pair occurs infinitely many times. This enumeration is tail-dense: for every compact set $K \subset \mathbb{R}^n$, every $G \in C(\mathbb{R}^n)$, every $\delta > 0$, and every $M \in \mathbb{N}$, there exists $m \geq M$ such that

$$K \subset Q_m \quad \text{and} \quad \|p_m - G\|_{C(K)} < \delta.$$

Indeed, one first chooses a rational compact box containing K , approximates G uniformly on that box by a polynomial, and then perturbs the coefficients slightly so that they are rational.

For each $m \geq 1$, by the compact-wise density assumption, choose $g_m \in \mathcal{N}_\varphi^\ell(\mathbb{R}^n)$ such that

$$\|g_m - p_m\|_{C(Q_m)} < 2^{-m}.$$

We now place these local targets far apart along the direction $\mathbf{b}$. Since $\mathbf{b} \neq 0$, the translates $Q_m + r\mathbf{b}$ escape every compact subset of $\mathbb{R}^n$ as $r \rightarrow \infty$. Choose integers

$$1 \leq n_1 < n_2 < \dots$$

inductively so that, with

$$E_m := Q_m + n_m \mathbf{b},$$

one has

$$\text{dist}(E_m, [-m, m]^n) > 2$$

and

$$\text{dist}(E_m, E_j) > 4 \quad \text{whenever } j < m.$$

For each m , set

$$U_m := \{\mathbf{x} \in \mathbb{R}^n : \text{dist}(\mathbf{x}, E_m) < 1\}.$$

Then the open sets U_m are pairwise disjoint and locally finite.

Choose $\chi_m \in C_c(\mathbb{R}^n)$ such that

$$0 \leq \chi_m \leq 1, \quad \chi_m \equiv 1 \text{ on } E_m, \quad \text{supp } \chi_m \subset U_m.$$

Define

$$h_m(\mathbf{x}) := \lambda^{-n_m} g_m(\mathbf{x} - n_m \mathbf{b}) \chi_m(\mathbf{x}),$$

and set

$$F(\mathbf{x}) := \sum_{m=1}^{\infty} h_m(\mathbf{x}).$$

Since the family $(U_m)_{m \geq 1}$ is locally finite, the above series is locally finite. Hence $F \in C(\mathbb{R}^n) = X_\ell^n$.

We claim that the orbit of F is dense. Fix $m \geq 1$ and $\mathbf{x} \in Q_m$. Then $\mathbf{x} + n_m \mathbf{b} \in E_m$. Therefore

$$\chi_m(\mathbf{x} + n_m \mathbf{b}) = 1.$$

Moreover, since the supports of the functions χ_j are contained in the pairwise disjoint sets U_j , no block with $j \neq m$ contributes at $\mathbf{x} + n_m \mathbf{b}$. Thus

$$\begin{aligned} (T_{\lambda, \mathbf{b}}^{n_m} F)(\mathbf{x}) &= \lambda^{n_m} F(\mathbf{x} + n_m \mathbf{b}) \\ &= \lambda^{n_m} h_m(\mathbf{x} + n_m \mathbf{b}) \\ &= \lambda^{n_m} \lambda^{-n_m} g_m((\mathbf{x} + n_m \mathbf{b}) - n_m \mathbf{b}) \chi_m(\mathbf{x} + n_m \mathbf{b}) \\ &= g_m(\mathbf{x}). \end{aligned}$$

Hence

$$(T_{\lambda, \mathbf{b}}^{n_m} F)|_{Q_m} = g_m|_{Q_m}.$$

Let now $G \in X_\ell^n = C(\mathbb{R}^n)$, let $K \subset \mathbb{R}^n$ be compact, and let $\varepsilon > 0$. Choose $M \in \mathbb{N}$ such that

$$2^{-M} < \frac{\varepsilon}{2}.$$

By tail-density, choose $m \geq M$ such that

$$K \subset Q_m \quad \text{and} \quad \|p_m - G\|_{C(K)} < \frac{\varepsilon}{2}.$$

Then

$$\begin{aligned} \|T_{\lambda, \mathbf{b}}^{n_m} F - G\|_{C(K)} &\leq \|T_{\lambda, \mathbf{b}}^{n_m} F - g_m\|_{C(K)} + \|g_m - p_m\|_{C(K)} + \|p_m - G\|_{C(K)} \\ &= 0 + \|g_m - p_m\|_{C(K)} + \|p_m - G\|_{C(K)} \\ &\leq \|g_m - p_m\|_{C(Q_m)} + \|p_m - G\|_{C(K)} \\ &< 2^{-m} + \frac{\varepsilon}{2} \\ &\leq 2^{-M} + \frac{\varepsilon}{2} < \varepsilon. \end{aligned}$$

Since basic neighborhoods in the compact-open topology are determined by finitely many compact sets, replacing those compact sets by their union gives density of the orbit of F . Therefore $T_{\lambda, \mathbf{b}}$ is hypercyclic on X_ℓ^n . $\square$

For a bounded set $S \subset \mathbb{R}^n$ and a unit vector $\mathbf{u}$, we write

$$w_{\mathbf{u}}(S) := \sup\{ |(\mathbf{x} - \mathbf{y}) \cdot \mathbf{u}| : \mathbf{x}, \mathbf{y} \in S \}$$

for the width of S in the direction $\mathbf{u}$.

Proposition 3.2. *let $\varphi \in C(\mathbb{R})$ be non-polynomial and $\lambda \neq 0$, $\mathbf{b} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$, $k \in \mathbb{N}$. Let*

$$\mathbf{u} := \frac{\mathbf{b}}{\|\mathbf{b}\|}.$$

Let $S \subset \mathbb{R}^n$ be a bounded set with nonempty interior such that

$$w_{\mathbf{u}}(S) < k\|\mathbf{b}\|.$$

Let $\psi \in C_c(\mathbb{R}^n)$ satisfy $\text{supp } \psi \subset S^\circ$, where S° denotes the interior of S in $\mathbb{R}^n$. Define

$$f(\mathbf{x}) := \sum_{j \in \mathbb{Z}} \lambda^{-kj} \psi(\mathbf{x} - jk\mathbf{b}).$$

Then $f \in X_\ell^n$ and $T_{\lambda, \mathbf{b}}^k f = f$. In particular, if $\psi \neq 0$, then f is a nonzero periodic point of $T_{\lambda, \mathbf{b}}$, with period dividing k .

Proof. First we show that the definition of f is meaningful. Since

$$w_{\mathbf{u}}(S) < k\|\mathbf{b}\|,$$

the translates

$$S + jk\mathbf{b}, \quad j \in \mathbb{Z},$$

are pairwise disjoint. Indeed, if

$$\mathbf{z} \in (S + jk\mathbf{b}) \cap (S + mk\mathbf{b})$$

for some $j \neq m$, then there exist $\mathbf{x}, \mathbf{y} \in S$ such that

$$\mathbf{z} = \mathbf{x} + jk\mathbf{b} = \mathbf{y} + mk\mathbf{b}.$$

Thus

$$\mathbf{x} - \mathbf{y} = (m - j)k\mathbf{b}.$$

Taking inner products with $\mathbf{u} = \mathbf{b}/\|\mathbf{b}\|$, we obtain

$$|(\mathbf{x} - \mathbf{y}) \cdot \mathbf{u}| = |m - j|k\|\mathbf{b}\| \geq k\|\mathbf{b}\|,$$

which contradicts

$$w_{\mathbf{u}}(S) < k\|\mathbf{b}\|.$$

Therefore the sets $S + jk\mathbf{b}$ are pairwise disjoint.

Since $\text{supp } \psi \subset S^\circ$, the translated supports

$$\text{supp } \psi + jk\mathbf{b}, \quad j \in \mathbb{Z},$$

are also pairwise disjoint. Moreover, because $\text{supp } \psi$ is compact and $\mathbf{b} \neq 0$, this family of supports is locally finite. Hence the series

$$f(\mathbf{x}) = \sum_{j \in \mathbb{Z}} \lambda^{-kj} \psi(\mathbf{x} - jk\mathbf{b})$$

is locally finite. Therefore f is well-defined and continuous on $\mathbb{R}^n$.

Next we prove the periodicity identity. For every $\mathbf{x} \in \mathbb{R}^n$,

$$\begin{aligned} (T_{\lambda, \mathbf{b}}^k f)(\mathbf{x}) &= \lambda^k f(\mathbf{x} + k\mathbf{b}) \\ &= \lambda^k \sum_{j \in \mathbb{Z}} \lambda^{-kj} \psi(\mathbf{x} + k\mathbf{b} - jk\mathbf{b}) \\ &= \lambda^k \sum_{j \in \mathbb{Z}} \lambda^{-kj} \psi(\mathbf{x} - (j - 1)k\mathbf{b}). \end{aligned}$$

Putting $i = j - 1$, we obtain

$$\begin{aligned} (T_{\lambda, \mathbf{b}}^k f)(\mathbf{x}) &= \lambda^k \sum_{i \in \mathbb{Z}} \lambda^{-k(i+1)} \psi(\mathbf{x} - ik\mathbf{b}) \\ &= \sum_{i \in \mathbb{Z}} \lambda^{-ki} \psi(\mathbf{x} - ik\mathbf{b}) \\ &= f(\mathbf{x}). \end{aligned}$$

Thus $T_{\lambda, \mathbf{b}}^k f = f$.

It remains to show that $f \in X_\ell^n$. Let $K \subset \mathbb{R}^n$ be compact and let $\varepsilon > 0$. Define

$$J(K) := \{j \in \mathbb{Z} : K \cap (\text{supp } \psi + jk\mathbf{b}) \neq \emptyset\}.$$

By local finiteness, $J(K)$ is finite. If $J(K) = \emptyset$, then $f = 0$ on K , and the conclusion is immediate because

$$0 \in \mathcal{N}_\varphi^\ell(\mathbb{R}^n).$$

Assume now that $J(K) \neq \emptyset$. Set

$$K' := \bigcup_{j \in J(K)} (K - jk\mathbf{b}).$$

Then K' is compact. By (3.1), choose $g \in \mathcal{N}_\varphi^\ell(\mathbb{R}^n)$ such that

$$\|\psi - g\|_{C(K')} < \delta,$$

where

$$\delta := \frac{\varepsilon}{\sum_{j \in J(K)} |\lambda|^{-kj}}.$$

By Lemma 3, the class $\mathcal{N}_\varphi^\ell(\mathbb{R}^n)$ is invariant under translations, and it is a linear space. Hence the function

$$G_K(\mathbf{x}) := \sum_{j \in J(K)} \lambda^{-kj} g(\mathbf{x} - jk\mathbf{b})$$

belongs to $\mathcal{N}_\varphi^\ell(\mathbb{R}^n)$.

For $\mathbf{x} \in K$, all nonzero terms of $f(\mathbf{x})$ correspond to indices $j \in J(K)$. Therefore

$$f(\mathbf{x}) = \sum_{j \in J(K)} \lambda^{-kj} \psi(\mathbf{x} - jk\mathbf{b}).$$

Consequently, for $\mathbf{x} \in K$,

$$\begin{aligned} |f(\mathbf{x}) - G_K(\mathbf{x})| &= \left| \sum_{j \in J(K)} \lambda^{-kj} (\psi(\mathbf{x} - jk\mathbf{b}) - g(\mathbf{x} - jk\mathbf{b})) \right| \\ &\leq \sum_{j \in J(K)} |\lambda|^{-kj} \|\psi - g\|_{C(K')} \\ &< \delta \sum_{j \in J(K)} |\lambda|^{-kj} \\ &= \varepsilon. \end{aligned}$$

Hence, $\|f - G_K\|_{C(K)} < \varepsilon$. Since K and ε were arbitrary, f belongs to the compact-open closure of $\mathcal{N}_\varphi^\ell(\mathbb{R}^n)$. So, $f \in X_\ell^n$. This completes the proof. $\square$

Now, using the same argument as in the proof of [Theorem 2.5](#) we can establish the next result.

Theorem 3.3. *let $\varphi \in C(\mathbb{R})$ be non-polynomial, and $\mathbf{b} \in \mathbb{R}^n \setminus \{0\}$, $\lambda \neq 0$, then $T_{\lambda,\mathbf{b}}$ is chaotic.*

4. Discussion

This work shows that universal approximation in the compact-open topology can be realized through the dynamics of a weighted translation operator acting on a single fixed seed. More precisely, for suitable activation functions, we provided an existence of a function $F \in X_1 := \overline{\mathcal{N}_\varphi^1(\mathbb{R})}^{co}$ such that its orbit under the operator

$$(T_{\lambda,\beta}f)(x) = \lambda f(x + \beta), \quad \lambda \neq 0, \quad \beta \neq 0,$$

is dense in X_1 for the compact-open topology. Since, under the compact universal approximation assumption (when φ is non-polynomial), we have $X_1 = C(\mathbb{R})$ with the compact-open topology, the orbit of this single seed approximates arbitrary continuous targets uniformly on compact sets. In particular, every target in $C_0(\mathbb{R})$ is approximable on compact subsets by suitable iterates of the same seed. Thus the approximation mechanism does not require re-optimizing the network parameters for each new target; instead, one moves along the orbit of a fixed function under a fixed linear operator.

We also identify a structural dynamical property behind this approximation mechanism. On each compact-open closure $X_\ell := \overline{\mathcal{N}_\varphi^\ell(\mathbb{R})}^{co}$ for which the corresponding network class is dense on compact sets, the weighted translation operator is hypercyclic. Moreover, periodic points are dense, and hence the operator is chaotic in the sense of linear dynamics. The periodic points are constructed by placing a compactly supported patch along a bi-infinite arithmetic progression in the translation direction. These periodic patterns belong to the compact-open closure of the relevant network classes because, on each compact set, only finitely many translated patches are visible and can be approximated by networks. Thus orbit-based approximation in this setting is driven by chaotic translation dynamics and by the geometry of spatially separated blocks.

The compact-open topology is essential for this construction. In a global uniform norm on $C_0(\mathbb{R})$, constant-weight translations cannot be hypercyclic: depending on the value of $|\lambda|$, the orbit either collapses toward zero, escapes in norm, or remains on a fixed norm sphere. By contrast, the compact-open topology only tests convergence on fixed compact sets. This allows one to place different approximation blocks farther and farther away in space, so that each block becomes visible only at a prescribed iterate. The construction therefore uses a separation in both space and time: different local targets are encoded into disjoint

translated regions and are read out by applying suitable powers of the same operator.

The role of $C_0(\mathbb{R})$ should be interpreted carefully. For arbitrary $\lambda \neq 0$, the hypercyclic vector constructed in the compact-open closure need not vanish at infinity. If one requires the hypercyclic seed itself to belong to $C_0(\mathbb{R})$, then the condition $|\lambda| > 1$ is needed in the one-sided block construction. This condition allows the amplitudes of the stored blocks to decay at infinity. On the other hand, the bi-infinite periodic-point construction does not require $|\lambda| > 1$; it works for every $\lambda \neq 0$, although nonzero periodic points cannot belong to $C_0(\mathbb{R})$.

The same orbit-based mechanism is stable under changes in depth and dimension. For every nonzero translation vector $\mathbf{b} \in \mathbb{R}^n \setminus \{0\}$ and every $\lambda \neq 0$, the operator

$$(T_{\lambda,\mathbf{b}}f)(\mathbf{x}) = \lambda f(\mathbf{x} + \mathbf{b})$$

is hypercyclic, and in fact chaotic, on X_ℓ^n . Thus the orbit-based approximation principle extends naturally across both depth and input dimension. In this sense, the expressive behavior comes not only from the network class itself, but also from the orbit structure generated by the translation dynamics.

Several directions remain open. From an analytical point of view, it would be interesting to classify other linear operators on spaces of continuous or locally integrable functions whose orbits are compatible with neural-network closures. Weighted translations provide a particularly transparent example because they allow the explicit placement and recovery of approximation blocks. A natural question is how far this block-placement method extends to other composition operators, weighted shifts, dilation operators, or operators arising from differential and integral equations.

Another important direction is to study orbit-based approximation in other topologies and function spaces. The compact-open topology is well suited to continuous functions on noncompact domains, while local L^p topologies may provide an analogous framework for Lebesgue spaces. In contrast, global uniform or global L^p norms are often too rigid for constant-weight translations. Understanding precisely which topologies are flexible enough to support hypercyclic orbit approximation, while still being strong enough for applications, remains an important problem.

The construction in this paper is qualitative: it proves the existence of hypercyclic vectors and dense periodic points. A central open problem is to construct such vectors explicitly and in a numerically stable way. In the ReLU case, one possible approach is to replace the abstract compact approximation step by explicit piecewise affine or basis-generated approximants.

The main numerical challenge is stability. A block stored at a far translate is rescaled by a factor involving λ^{-n} , and the corresponding read-out under $T_{\lambda,\beta}^n$ amplifies perturbations by approximately $|\lambda|^n$. Thus the mathematical orbit interpolation is exact, but its numerical implementation may be ill-conditioned when $|\lambda| > 1$ and the iterate n is large. This suggests a distinction between infinite hypercyclic seeds, which are useful for universality, and finite task-specific seeds, which may be more appropriate for computation.

From an algorithmic perspective, several questions arise. Can one design training procedures that approximate a hypercyclic seed? How robust are such seeds under perturbations, discretization, and numerical noise? How should the iterate index n be selected or learned for a given target function? Can one combine orbit-based approximation with existing optimization methods so that part of the approximation power comes from the operator dynamics rather than from retraining all parameters?

Finally, the activation assumptions deserve further clarification. The compact-open results in this paper primarily require compact-wise density of the relevant network class. For one-hidden-layer networks, this is guaranteed by the classical universal approximation theorem when $\varphi \in C(\mathbb{R})$ is non-polynomial. Additional assumptions, such as asymptotic linearity, may be relevant for noncompact approximation in stronger

norms, but they are not essential for the compact–open hypercyclicity mechanism itself. It would be valuable to identify the weakest activation conditions under which orbit-based universality, chaoticity, and possible C_0 -valued constructions remain valid.

Overall, the results suggest a dynamical viewpoint on neural network approximation: instead of approximating each target by choosing a new network, one may encode a dense family of local targets into a single seed and recover them by iterating a fixed operator. This perspective opens several theoretical and computational questions, particularly concerning explicit constructions, stability, quantitative rates, and applications to families of integral or differential equations.

CRedit authorship contribution statement

Farzona Mukhamedova: Writing – review & editing, Writing – original draft, Validation, Resources, Project administration, Methodology, Investigation, Data curation, Conceptualization; **Ivan Tyukin:** Writing – original draft, Supervision, Methodology, Conceptualization; **Farrukh Mukhamedov:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization.

Ethical Approval

This study constitutes theoretical research and does not involve human participants, animal subjects, or biological materials. Therefore, no ethical approval was required.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors thank the reviewers for their constructive and valuable suggestions which helped improve the presentation of the paper.

Data Availability Statement

No data is used in the present work.

References

- Asano, M., Ohya, M., Tanaka, Y., Basieva, I., & Khrennikov, A. (2011). Quantum-like model of brain's functioning: Decision making from decoherence. *Journal of Theoretical Biology*, 281(1), 56–64.
- Barron, A. (1991). Universal approximation bounds for superposition of a sigmoidal function. *IEEE Transaction on Information Theory*, 19, 930–944.
- Barron, A. R. (2002). Universal approximation bounds for superpositions of a sigmoidal function. *IEEE Transactions on Information Theory*, 39(3), 930–945.
- Bayart, F., & Étienne Matheron, M. (2009). Dynamics of linear operators. In *Number 179 in cambridge tracts in mathematics*. Cambridge university press.
- Cybenko, G. (1989). Approximation by superpositions of a sigmoidal function. *Mathematics of Control, Signals and Systems*, 2(4), 303–314.
- Eldan, R., & Shamir, O. (2016). The power of depth for feedforward neural networks. In *Conference on learning theory* (pp. 907–940).
- Funahashi, K.-I., & Nakamura, Y. (1993). Approximation of dynamical systems by continuous time recurrent neural networks. *Neural Networks*, 6(6), 801–806.
- Guliyev, N. J., & Ismailov, V. E. (2018). On the approximation by single hidden layer feedforward neural networks with fixed weights. *Neural Networks*, 98, 296–304.
- Gorban, A. N., Tyukin, I. Y., Prokhorov, D. V., & Sofeikov, K. I. (2016). Approximation with random bases: Pro-et contra. *Information Sciences*, 364, 129–145.
- Grosse-Erdmann, K.-G., & Manguillot, A. P. (2011). *Linear chaos* Springer Science & Business Media.
- Hanin, B., & Rolnick, D. (2019). Complexity of linear regions in deep networks. In *International conference on machine learning* (pp. 2596–2604).
- Hanin, B., & Sellke, M. (2017). Approximating continuous functions by relu nets of minimal width. *arxiv:1710.11278*.
- Hornik, K. (1991). Approximation capabilities of multilayer feedforward networks. *Neural Networks*, 4(2), 251–257.
- Hornik, K., Stinchcombe, M., & White, H. (1989). Multilayer feedforward networks are universal approximators. *Neural Networks*, 2(5), 359–366.
- Khrennikov, A., Basieva, I., Pothos, E. M., & Yamato, I. (2018). Quantum probability in decision making from quantum information representation of neuronal states. *Scientific Reports*, 8(1), 16225.
- Khrennikov, A., Ozawa, M., Benninger, F., & Shor, O. (2025). Coupling quantum-like cognition with the neuronal networks within generalized probability theory. *Journal of Mathematical Psychology*, 125, 102923.
- Kolmogorov, A. N. (1957). On the representation of continuous functions of many variables by superposition of continuous functions of one variable and addition. *Doklady Akademii Nauk SSSR*, 114(5), 953–956. <http://mi.mathnet.ru/dan22050>. MR0111809, Zbl 0090.27103.
- Leshno, M., Lin, V. Y., Pinkus, A., & Schocken, S. (1993). Multilayer feedforward networks with a nonpolynomial activation function can approximate any function. *Neural Networks*, 6(6), 861–867.
- Liang, S., & Srikant, R. (2016). Why deep neural networks for function approximation? *arxiv:1610.04161*.
- Lin, H. W., Tegmark, M., & Rolnick, D. (2017). Why does deep and cheap learning work so well. *Journal of Statistical Physics*, 168(6), 1223–1247.
- Long, J., Wu, W., & Nan, D. (2007). Uniform approximation capabilities of sum-of-product and sigma-pi-sigma neural networks. In *International symposium on neural networks* (pp. 1110–1116). Springer.
- Lu, Z., Pu, H., Wang, F., Hu, Z., & Wang, L. (2017). The expressive power of neural networks: A view from the width. *Advances in Neural Information Processing Systems*, 30.
- Maierov, V., & Pinkus, A. (1999). Lower bounds for approximation by mlp neural networks. *Neurocomputing*, 25(1–3), 81–91.
- Michael, E. (1961). On a theorem of rudin and klee. *Proceedings of the AMS*, 12, 921.
- Montúfar, G., Pascanu, R., Cho, K., & Bengio, Y. (2014). On the number of linear regions of deep neural networks. *Advances in Neural Information Processing Systems*, 27.
- Mukhamedov, F., & Khakimov, O. (2018). Hypercyclic and supercyclic linear operators on non-archimedean vector spaces. *The Bulletin of the Belgian Mathematical Society*, 25(1), 85–105.
- Mukhamedov, F., Khakimov, O., & Souissi, A. (2022). A few remarks on supercyclicity of non-archimedean linear operators on $c_0(\mathbb{N})$. *p-adic Numb Ultra. Analysis and Applications*, 14(22), 63–75.
- Petersen, P., & Voigtlaender, F. (2018). Optimal approximation of piecewise smooth functions using deep relu neural networks. *Neural Networks*, 108, 296–330.
- Pinkus, A. (1999). Approximation theory of the mlp model in neural networks. *Acta Numerica*, 8, 143–195.
- Raghu, M., Poole, B., Kleinberg, J., Ganguli, S., & Sohl-Dickstein, J. (2017). On the expressive power of deep neural networks. In *International conference on machine learning* PMLR (pp. 2847–2854).
- Stone, M. H., (1948). The generalized weierstrass approximation theorem. *Mathematics Magazine*, 21(5), 237–254.
- Telgarsky, M. (2016). Benefits of depth in neural networks. In *Conference on learning theory* (pp. 1517–1539).
- Teun, D. H., & Nuland, V. (2024). Noncompact uniform universal approximation. *Neural Networks*, 173, 106181.
- Tyukin, I. Y., Grechuk, B., Evgeny, E. M., & Gorban, A. N. (2025). When fractional quasi p-norms concentrate. *arxiv:2505.19635*.
- Weierstrass, K. (1885). Über die analytische darstellbarkeit sogenannter willkürlicher funktionen reeller argumente.
- Yarotsky, D. (2017). Error bounds for approximations with deep relu networks. *Neural Networks*, 94, 103–114.